# BANK MANAGEMENT SYSTEM

PROJECT OVERVIEW :

The Bank Management Application is a mini project developed in C++ that helps in managing banking activities in a simple and organized way. The application provides a menu-driven interface through which users can perform different banking operations easily.

This project is developed using Object-Oriented Programming (OOP) concepts such as classes, objects, and functions. It also uses file handling to store customer account details permanently.

The application allows users to:

Open a new bank account

Deposit money into an account

Withdraw money from an account

View account details and balance

Maintain customer records digitally

The project reduces manual calculations and improves accuracy in maintaining banking transactions. It is mainly designed for educational purposes to understand the practical implementation of C++ programming concepts.

FEATURES

Account Creation

Money Deposit

Money Withdrawal

Balance Inquiry

Data Storage Using Files

Simple User Interface

TECHNOLOGIES USED

C++

File Handling (fstream)

Vectors

Functions

Classes and Objects

# BANK MANAGEMENT SYSTEM

OBJECTIVES

To develop a simple banking system

To manage customer account information efficiently

To perform secure banking transactions

To understand real-time application development using C++

CONCLUSION

The Bank Management Application is an efficient and user-friendly project that performs essential banking operations digitally. It provides practical knowledge of programming concepts and demonstrates how C++ can be used to build real-world applications.

CODE :

```
#include <iostream>

#include <fstream>

#include <vector>

#include <iomanip>

using namespace std;


class Bank {

    string accNo, name;

    double balance;


public:

    Bank(string a, string n, double b) {

        accNo = a;

        name = n;

        balance = b;
```

```
}

string getAccNo() { return accNo; }

void deposit(double amt) {
    if (amt > 0) {
        balance += amt;
        cout << "\nDeposit Successful!\nBalance: $"
             << fixed << setprecision(2) << balance << endl;
    }
}

void withdraw(double amt) {
    if (amt <= balance && amt > 0) {
        balance -= amt;
        cout << "\nWithdrawal Successful!\nBalance: $"
             << fixed << setprecision(2) << balance << endl;
    } else {
        cout << "\nInsufficient Balance!" << endl;
    }
}

void display() {
    cout << "\nAccount Number : " << accNo;
    cout << "\nAccount Holder : " << name;
    cout << "\nBalance        : $"
```

```
        << fixed << setprecision(2) << balance << endl;
    }

    void save(ofstream &out) {
        out << accNo << " " << name << " " << balance << endl;
    }
};

void saveFile(vector<Bank> &acc) {
    ofstream out("bank.txt");
    for (auto &a : acc)
        a.save(out);
}

int main() {
    vector<Bank> acc;
    int ch;
    string no, name;
    double amt;

    do {
        cout << "\n====== BANK MANAGEMENT SYSTEM ======";
        cout << "\n1.Create Account";
        cout << "\n2.Deposit";
        cout << "\n3.Withdraw";
        cout << "\n4.Check Balance";
```

```
cout << "\n5.Exit";
cout << "\nEnter choice: ";
cin >> ch;

switch (ch) {

case 1:
   cout << "\nEnter Account Number: ";
   cin >> no;

   cout << "Enter Name: ";
   cin >> ws;
   getline(cin, name);

   cout << "Enter Balance: ";
   cin >> amt;

   acc.push_back(Bank(no, name, amt));
   saveFile(acc);

   cout << "\nAccount Created Successfully!" << endl;
   break;

case 2:
   cout << "\nEnter Account Number: ";
   cin >> no;
```

```
        for (auto &a : acc) {
            if (a.getAccNo() == no) {
                cout << "Enter Deposit Amount: ";
                cin >> amt;
                a.deposit(amt);
                saveFile(acc);
            }
        }
        break;

    case 3:
        cout << "\nEnter Account Number: ";
        cin >> no;

        for (auto &a : acc) {
            if (a.getAccNo() == no) {
                cout << "Enter Withdraw Amount: ";
                cin >> amt;
                a.withdraw(amt);
                saveFile(acc);
            }
        }
        break;

    case 4:
```

```
        cout << "\nEnter Account Number: ";
        cin >> no;

        for (auto &a : acc) {
            if (a.getAccNo() == no)
                a.display();
        }
        break;

    case 5:
        cout << "\nThank You!" << endl;
        break;

    default:
        cout << "\nInvalid Choice!" << endl;
    }

} while (ch != 5);

return 0;
}
```

SAMPLE OUTPUT :

====== BANK MANAGEMENT SYSTEM ======

1.Create Account

2.Deposit

3.Withdraw

4.Check Balance

5.Exit

Enter choice: 1

Enter Account Number: 533006

Enter Name: Ram

Enter Balance: 20000

Account Created Successfully!

====== BANK MANAGEMENT SYSTEM ======

1.Create Account

2.Deposit

3.Withdraw

4.Check Balance

5.Exit

Enter choice: 3

Enter Account Number: 533006

Enter Withdraw Amount: 15000

Withdrawal Successful!

Balance: $5000.00

====== BANK MANAGEMENT SYSTEM ======

# BANK MANAGEMENT SYSTEM

1.Create Account

2.Deposit

3.Withdraw

4.Check Balance

5.Exit

Enter choice: 4

Enter Account Number: 533006

Account Number : 533006

Account Holder : Ram

Balance        : $5000.00

====== BANK MANAGEMENT SYSTEM ======

1.Create Account

2.Deposit

3.Withdraw

4.Check Balance

5.Exit

Enter choice: 5

Thank You!